KAIZEN MATHS

# Lesson starter

**Advanced Matrices: Transformations and Eigenvalues**

Cayley-Hamilton and Powers

Name

**S1**

Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 1 & -3 \\ 0 & 4 \end{bmatrix}$

**S2**

Use Cayley-Hamilton to find the inverse of each matrix.

$A = \begin{bmatrix} 4 & 2 \\ 0 & 5 \end{bmatrix}$

**S3**

Find the stated power of each matrix using its diagonalised form.

$A = \begin{bmatrix} 4 & 2 \\ 0 & 6 \end{bmatrix}, n = 2$

**S4**

Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 3 & -2 \\ 0 & 5 \end{bmatrix}$



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KAIZEN MATHS

# Exit ticket

**Advanced Matrices: Transformations and Eigenvalues**

Cayley-Hamilton and Powers

Name _____

**EA1**

Use Cayley-Hamilton to find the inverse of each matrix.

$A = \begin{bmatrix} 5 & 1 \\ 0 & 4 \end{bmatrix}$

**EA2**

Find the stated power of each matrix using its diagonalised form.

$A = \begin{bmatrix} 3 & 2 \\ 0 & 5 \end{bmatrix}, n = 2$



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# Exit ticket

**Advanced Matrices: Transformations and Eigenvalues**

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Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 2 & 4 \\ 0 & 4 \end{bmatrix}$



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# Exit ticket

**Advanced Matrices: Transformations and Eigenvalues**

Cayley-Hamilton and Powers

Name _____

**EA1**

Use Cayley-Hamilton to find the inverse of each matrix.

$A = \begin{bmatrix} 5 & 1 \\ 0 & 4 \end{bmatrix}$

**EA2**

Find the stated power of each matrix using its diagonalised form.

$A = \begin{bmatrix} 3 & 2 \\ 0 & 5 \end{bmatrix}, n = 2$

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KAIZEN MATHS

# Exit ticket

**Advanced Matrices: Transformations and Eigenvalues**

Cayley-Hamilton and Powers

Name _____

**EB1**

Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 2 & 4 \\ 0 & 4 \end{bmatrix}$

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# Teacher answers & working

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## S1

Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 1 & -3 \\ 0 & 4 \end{bmatrix}$

**Answer:**

$A^2 - 5A + 4I = 0$

**Working:**

The characteristic equation is $\lambda^2 - 5\lambda + 4 = 0$.

By the Cayley-Hamilton theorem, the matrix A satisfies its own characteristic equation.

Replace $\lambda$ by A.

$A^2 - 5A + 4I = 0$.

## S2

Use Cayley-Hamilton to find the inverse of each matrix.

$A = \begin{bmatrix} 4 & 2 \\ 0 & 5 \end{bmatrix}$

**Answer:**

$\begin{bmatrix} \frac{1}{4} & \frac{-1}{10} \\ 0 & \frac{1}{5} \end{bmatrix}$

**Working:**

The characteristic equation is $\lambda^2 - 9\lambda + 20 = 0$.

By Cayley-Hamilton: $A^2 - 9A + 20I = 0$.

Multiply by $A^{-1}$: $A - 9I + 20A^{-1} = 0$.

Rearrange: $20A^{-1} = 9I - A$.

So $A^{-1} = \frac{1}{20}(9I - A)$.

$9I - A = \begin{bmatrix} 5 & -2 \\ 0 & 4 \end{bmatrix}$.

Therefore $A^{-1} = \begin{bmatrix} \frac{1}{4} & \frac{-1}{10} \\ 0 & \frac{1}{5} \end{bmatrix}$.

# Teacher answers & working

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## S3

Find the stated power of each matrix using its diagonalised form.

$A = \begin{bmatrix} 4 & 2 \\ 0 & 6 \end{bmatrix}$, $n = 2$

**Answer:**

$\begin{bmatrix} 4^2 & 6^2 - 4^2 \\ 0 & 6^2 \end{bmatrix}$

**Working:**

This matrix can be written as $A = PDP^{-1}$.

Use $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}$.

Then $A^2 = PD^2P^{-1}$.

$D^2 = \begin{bmatrix} 4^2 & 0 \\ 0 & 6^2 \end{bmatrix}$.

For this form, $A^2 = \begin{bmatrix} 4^2 & 6^2 - 4^2 \\ 0 & 6^2 \end{bmatrix}$.

## S4

Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 3 & -2 \\ 0 & 5 \end{bmatrix}$

**Answer:**

$A^2 - 8A + 15I = 0$

**Working:**

The characteristic equation is $\lambda^2 - 8\lambda + 15 = 0$.

By the Cayley-Hamilton theorem, the matrix A satisfies its own characteristic equation.

Replace $\lambda$ by A.

$A^2 - 8A + 15I = 0$.

# Teacher answers & working

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## EA1

Use Cayley-Hamilton to find the inverse of each matrix.

$A = \begin{bmatrix} 5 & 1 \\ 0 & 4 \end{bmatrix}$

**Answer:**

$\begin{bmatrix} \frac{1}{5} & \frac{-1}{20} \\ 0 & \frac{1}{4} \end{bmatrix}$

**Working:**

The characteristic equation is $\lambda^2 - 9\lambda + 20 = 0$.

By Cayley-Hamilton: $A^2 - 9A + 20I = 0$.

Multiply by $A^{-1}$: $A - 9I + 20A^{-1} = 0$.

Rearrange: $20A^{-1} = 9I - A$.

So $A^{-1} = \frac{1}{20}(9I - A)$.

$9I - A = \begin{bmatrix} 4 & -1 \\ 0 & 5 \end{bmatrix}$.

Therefore $A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{-1}{20} \\ 0 & \frac{1}{4} \end{bmatrix}$.

## EA2

Find the stated power of each matrix using its diagonalised form.

$A = \begin{bmatrix} 3 & 2 \\ 0 & 5 \end{bmatrix}, n = 2$

**Answer:**

$\begin{bmatrix} 3^2 & 5^2 - 3^2 \\ 0 & 5^2 \end{bmatrix}$

**Working:**

This matrix can be written as $A = PDP^{-1}$.

Use $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$.

Then $A^2 = PD^2P^{-1}$.

$D^2 = \begin{bmatrix} 3^2 & 0 \\ 0 & 5^2 \end{bmatrix}$.

For this form, $A^2 = \begin{bmatrix} 3^2 & 5^2 - 3^2 \\ 0 & 5^2 \end{bmatrix}$.

# Teacher answers & working

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## EB1

Use the characteristic equation to write a Cayley-Hamilton relation for each matrix.

$A = \begin{bmatrix} 2 & 4 \\ 0 & 4 \end{bmatrix}$

**Answer:**

$A^2 - 6A + 8I = 0$

**Working:**

The characteristic equation is $\lambda^2 - 6\lambda + 8 = 0$.

By the Cayley-Hamilton theorem, the matrix A satisfies its own characteristic equation.

Replace $\lambda$ by A.

$A^2 - 6A + 8I = 0$.